Task 2: Data Retrieval & Sorting Dashboard
Description: Create a page that displays student or employee records with Sorting by name or date Filtering by department Count of students per departments

Algorithm

1. Start
2. Initialize employee data
3. Create a list (array) of employee records
4. Each record contains:
- ID
- Name
- Department
- Joining Date

5.Display all records
6.Load all employee data into a table on the webpage
7.Create department filter
8.Provide a dropdown with department options (IT, HR, Finance, All)
9.Filter records
10.If user selects a department:
- Show only employees belonging to that department
- If "All" is selected:
- Display all records
- Sort by name

11.When user clicks “Sort by Name”:
- Arrange records in ascending alphabetical order
- Sort by date

12.When user clicks “Sort by Date”:
- Arrange records based on joining date
- Count employees per department
- Traverse all employee records
- Count number of employees in each department
- Display department count
- Show the count below the table
- Update dynamically

13.Whenever filtering or sorting is applied:
- Refresh table data
- Update department count

14 End

Programs

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Employee Dashboard</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

   <div class="container">
```

```html
        <h1>Employee Dashboard</h1>

        <div class="controls">
            <label for="departmentFilter">Filter by Department:</label>
            <select id="departmentFilter">
                <option value="All">All</option>
                <option value="IT">IT</option>
                <option value="HR">HR</option>
                <option value="Finance">Finance</option>
            </select>

            <button onclick="sortByName()">Sort by Name</button>
            <button onclick="sortByDate()">Sort by Date</button>
        </div>

        <h2>Employee Records</h2>
        <table>
            <thead>
                <tr>
                    <th>ID</th>
                    <th>Name</th>
                    <th>Department</th>
                    <th>Joining Date</th>
                </tr>
            </thead>
            <tbody id="employeeTableBody"></tbody>
        </table>

        <h2>Department Count</h2>
        <ul id="departmentCount"></ul>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

Style.css

```css
body {
    font-family: Arial, sans-serif;
    background-color: #f4f4f4;
    margin: 0;
    padding: 20px;
}

.container {
    max-width: 900px;
    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
}

h1, h2 {
    text-align: center;
}
```

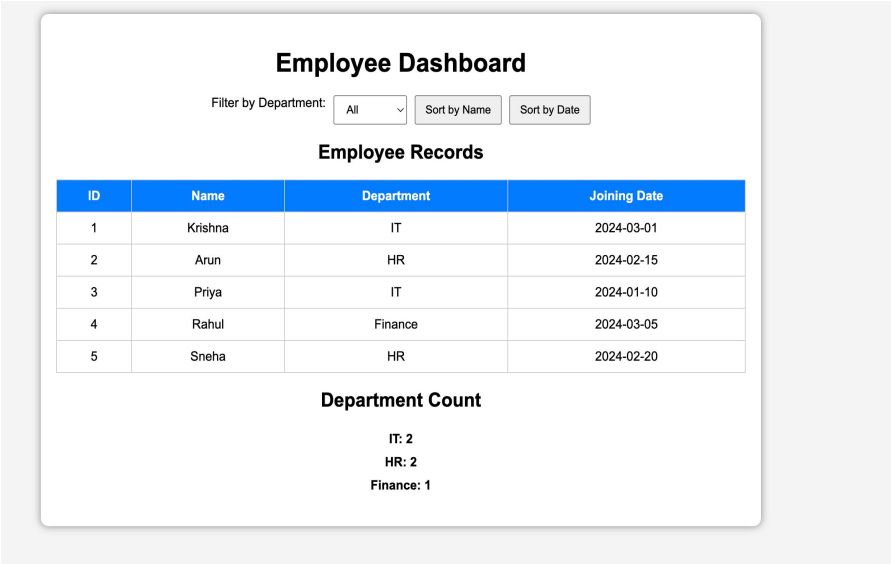
```
.controls {
    display: flex;
    gap: 10px;
    justify-content: center;
    margin-bottom: 20px;
    flex-wrap: wrap;
}

select, button {
    padding: 8px 12px;
    font-size: 14px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
}

table, th, td {
    border: 1px solid #ccc;
}

th, td {
    padding: 10px;
    text-align: center;
}

th {
    background-color: #007bff;
    color: white;
}

ul {
    list-style-type: none;
    padding: 0;
    text-align: center;
}

li {
    padding: 5px;
    font-weight: bold;
}
```

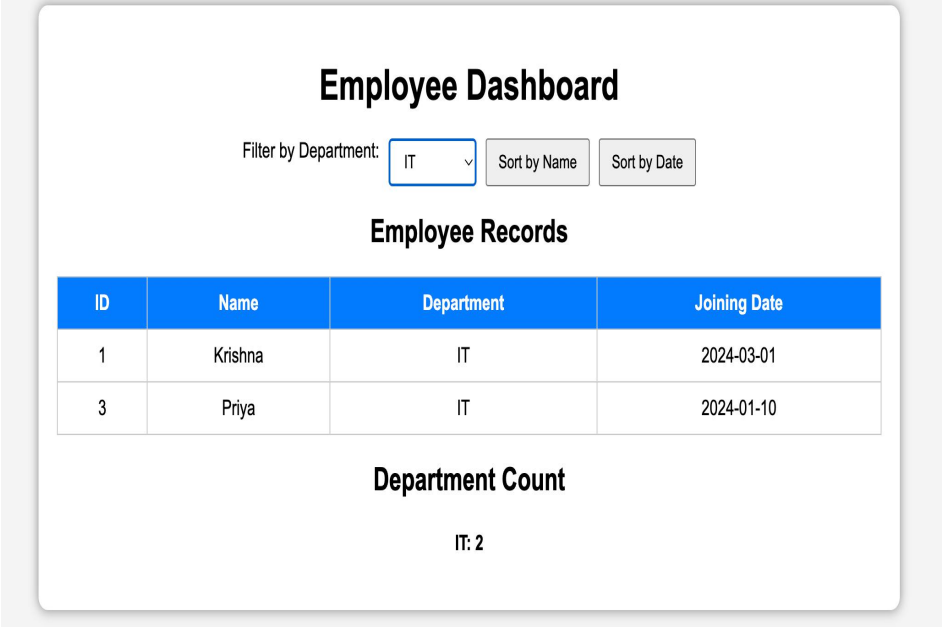
Script.js

```
const employees = [
    { id: 1, name: "Krishna", department: "IT", date: "2024-03-01" },
    { id: 2, name: "Arun", department: "HR", date: "2024-02-15" },
    { id: 3, name: "Priya", department: "IT", date: "2024-01-10" },
    { id: 4, name: "Rahul", department: "Finance", date: "2024-03-05" },
    { id: 5, name: "Sneha", department: "HR", date: "2024-02-20" }
];

let displayedEmployees = [...employees];

function displayEmployees(data) {
    const tableBody = document.getElementById("employeeTableBody");
```

```
    tableBody.innerHTML = "";

    data.forEach(emp => {
        const row = `
            <tr>
                <td>${emp.id}</td>
                <td>${emp.name}</td>
                <td>${emp.department}</td>
                <td>${emp.date}</td>
            </tr>
        `;
        tableBody.innerHTML += row;
    });
}

function updateDepartmentCount(data) {
    const countList = document.getElementById("departmentCount");
    countList.innerHTML = "";

    const counts = {};

    data.forEach(emp => {
        counts[emp.department] = (counts[emp.department] || 0) + 1;
    });

    for (let dept in counts) {
        const li = document.createElement("li");
        li.textContent = `${dept}: ${counts[dept]}`;
        countList.appendChild(li);
    }
}

function filterByDepartment() {
    const selectedDept = document.getElementById("departmentFilter").value;

    if (selectedDept === "All") {
        displayedEmployees = [...employees];
    } else {
        displayedEmployees = employees.filter(emp => emp.department === selectedDept);
    }

    displayEmployees(displayedEmployees);
    updateDepartmentCount(displayedEmployees);
}

function sortByName() {
    displayedEmployees.sort((a, b) => a.name.localeCompare(b.name));
    displayEmployees(displayedEmployees);
}

function sortByDate() {
    displayedEmployees.sort((a, b) => new Date(a.date) - new Date(b.date));
    displayEmployees(displayedEmployees);
}

document.getElementById("departmentFilter").addEventListener("change",
filterByDepartment);
```

```
displayEmployees(displayedEmployees);
updateDepartmentCount(displayedEmployees);
```





Employee Dashboard

Filter by Department: All | Sort by Name | Sort by Date

Employee Records

ID	Name	Department	Joining Date
2	Arun	HR	2024-02-15
1	Krishna	IT	2024-03-01
3	Priya	IT	2024-01-10
4	Rahul	Finance	2024-03-05
5	Sneha	HR	2024-02-20

Department Count

IT: 2

HR: 2

Finance: 1

Employee Dashboard

Filter by Department: All | Sort by Name | Sort by Date

Employee Records

ID	Name	Department	Joining Date
3	Priya	IT	2024-01-10
2	Arun	HR	2024-02-15
5	Sneha	HR	2024-02-20
1	Krishna	IT	2024-03-01
4	Rahul	Finance	2024-03-05

Department Count

IT: 2

HR: 2

Finance: 1

Result:Thus is an Data Retrieval & Sorting Dashboard has been done successfully